Report

2026-10-19

Part I Last week

1 Estimators

Estimated motion gain

$$\hat{M}[k] = \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]$$
$$\hat{\bar{a}}[k+1] = \hat{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\hat{M}[k] + \ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]$$

$\bar{a}'$ as a state variable

$$\hat{\bar{a}}'[k+1] = \hat{\bar{a}}'[k] + \ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]$$
$$q_{55}[k] = p_{\text{fix}}^2\,I[k]$$
$$\sqrt{p_{\text{fix}}} = 0.4\left(1.1/h_d[k]\right)^2$$

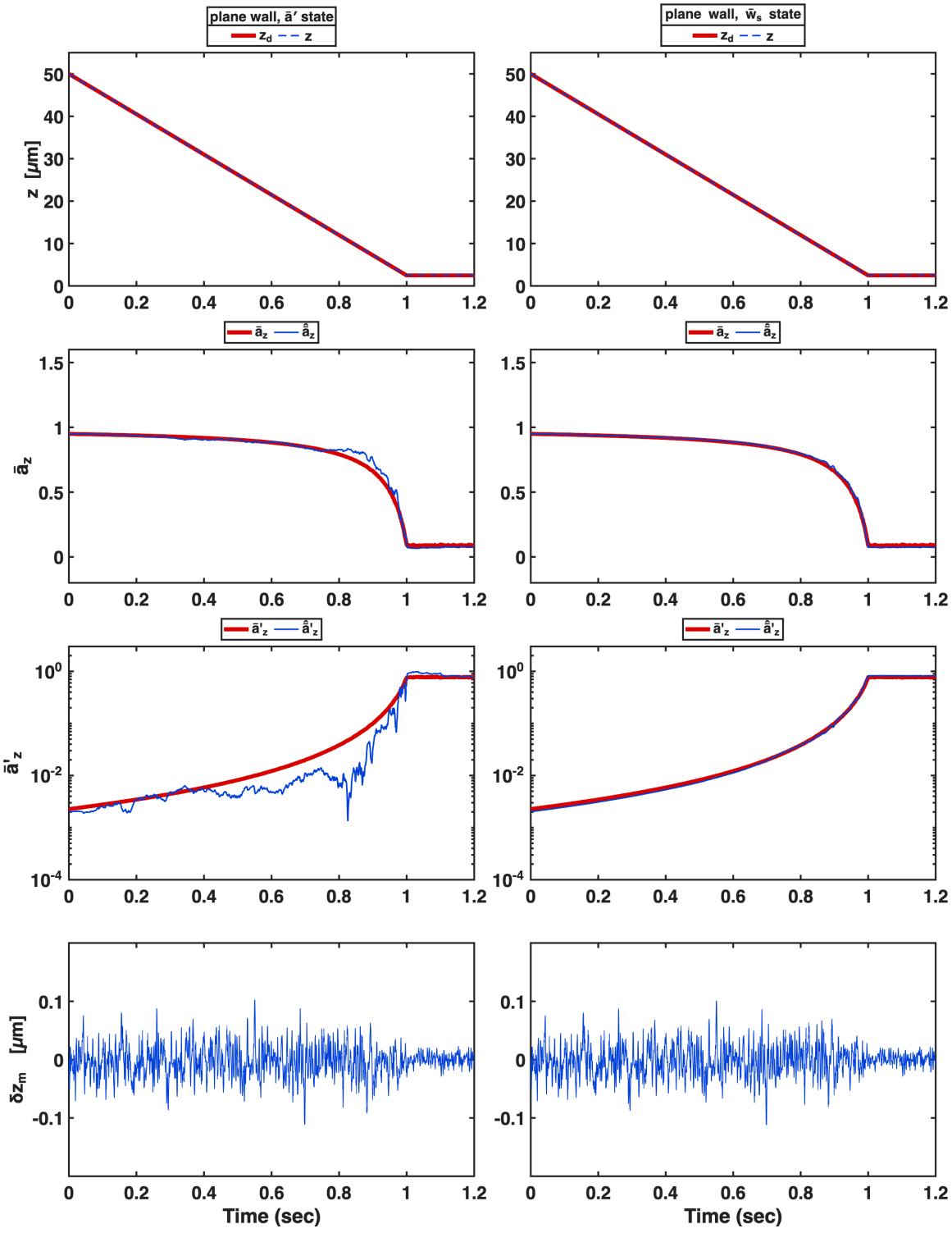
Wall position $\bar{w}_s$ as a state variable

$$\hat{h}[k] = \bar{x}_d[k] - \delta\hat{\bar{x}}_3[k] - \hat{\bar{w}}_s[k]$$
$$\hat{\bar{a}}'[k] = \frac{1}{\hat{h}[k]^2}$$
$$\hat{\bar{w}}_s[k+1] = \hat{\bar{w}}_s[k] + \ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]$$
$$q_{55} = 0$$

2 Motion control

Fig. 2.1 Constant-velocity trajectory on the plane wall

z axis; 22.2 → 1.10 R in 1 s, then held 0.2 s; plane wall; left: $\bar{a}'$ as a state variable; right: wall position $\bar{w}_s$ as a state variable, $\hat{\bar{w}}_s[0] = 0$; rows: z_d and z, $\bar{a}_z$, $\bar{a}'_z$ (log scale, right column $1/\hat{h}^2$), δz_m; $\lambda_c = 0.4$, seed 1



3 Regulation of the motion gain

Regulation law

$$\Delta\bar{x}_d[k] = (1-\lambda_a)\,\frac{\bar{a}_d - \hat{\bar{a}}[k]}{\hat{\bar{a}}'[k]}$$

$$\bar{x}_d[k+1] = \bar{x}_d[k] + \Delta\bar{x}_d[k]$$

$$\lambda_a = 1 - T_s/\tau_a, \qquad \tau_a = 0.1\,\mathrm{s}$$

$\bar{a}'$ as a state variable: limit on $\Delta\bar{x}_d$

$$\left|\Delta\bar{x}_d[k]\right| \le v_{\max}$$

$$v_{\max} = (22.2 - 1.10)\,T_s/1\,\mathrm{s}$$

Fig. 3.1 Regulation on the plane wall

z axis; start parked at 22.2 R, run 6 s; $\bar{a}_d = 0.07$; left: $\bar{a}'$ as a state variable, $|\Delta\bar{x}_d| \leq v_{\max}$; right: wall position $\bar{w}_s$ as a state variable; rows: z_d, z and z^* (height where $\bar{a} = \bar{a}_d$), $\bar{a}_z$, $\bar{a}'_z$ (log scale), δz_m; $\lambda_c = 0.4$, seed 1

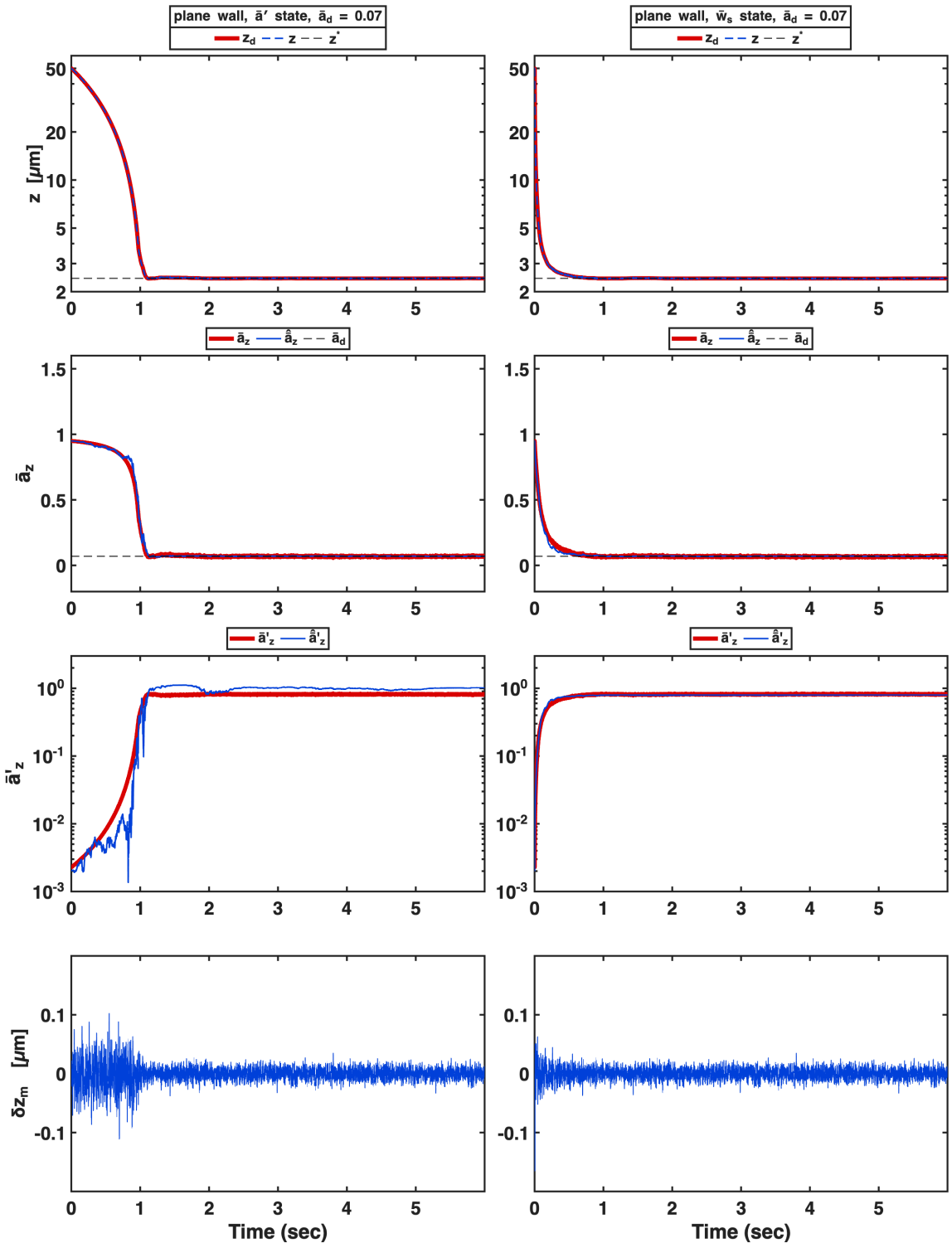
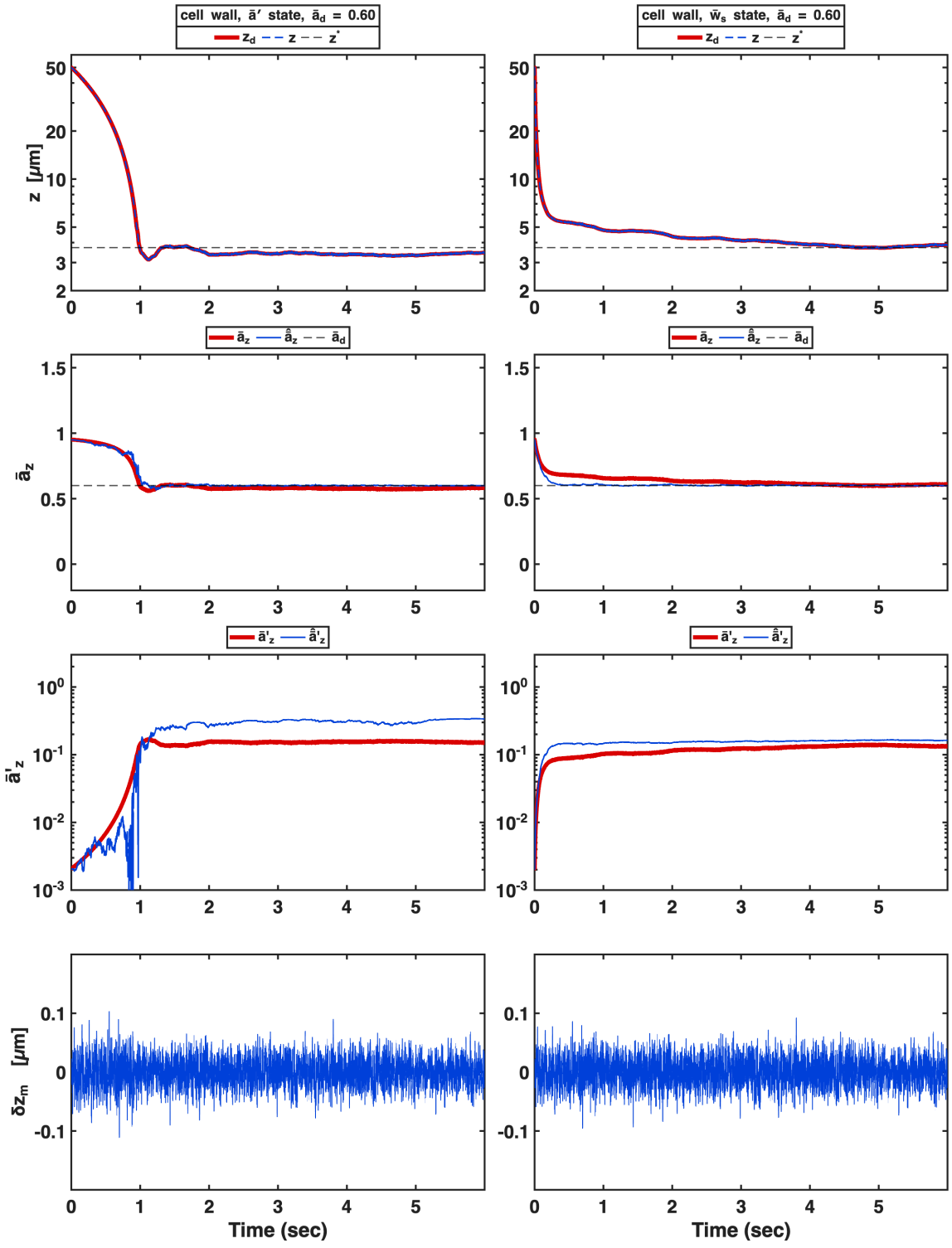


Fig. 3.2 Regulation on the cell wall

z axis; start parked at 22.2 R, run 6 s; $\bar{a}_d = 0.60$; cell wall = plane polynomial slid by 1.05 R, plant only; estimators started from the plane; left: $\bar{a}'$ as a state variable, $|\Delta\bar{x}_d| \leq v_{\max}$; right: wall position $\bar{w}_s$ as a state variable; rows as in Fig. 3.1; $\lambda_c = 0.4$, seed 1



Part II $\bar{a}'$ as a state variable

4 Measured input gain as the second output

Two choices of the second output

$$y_2[k] = \bar{a}_m[k]$$

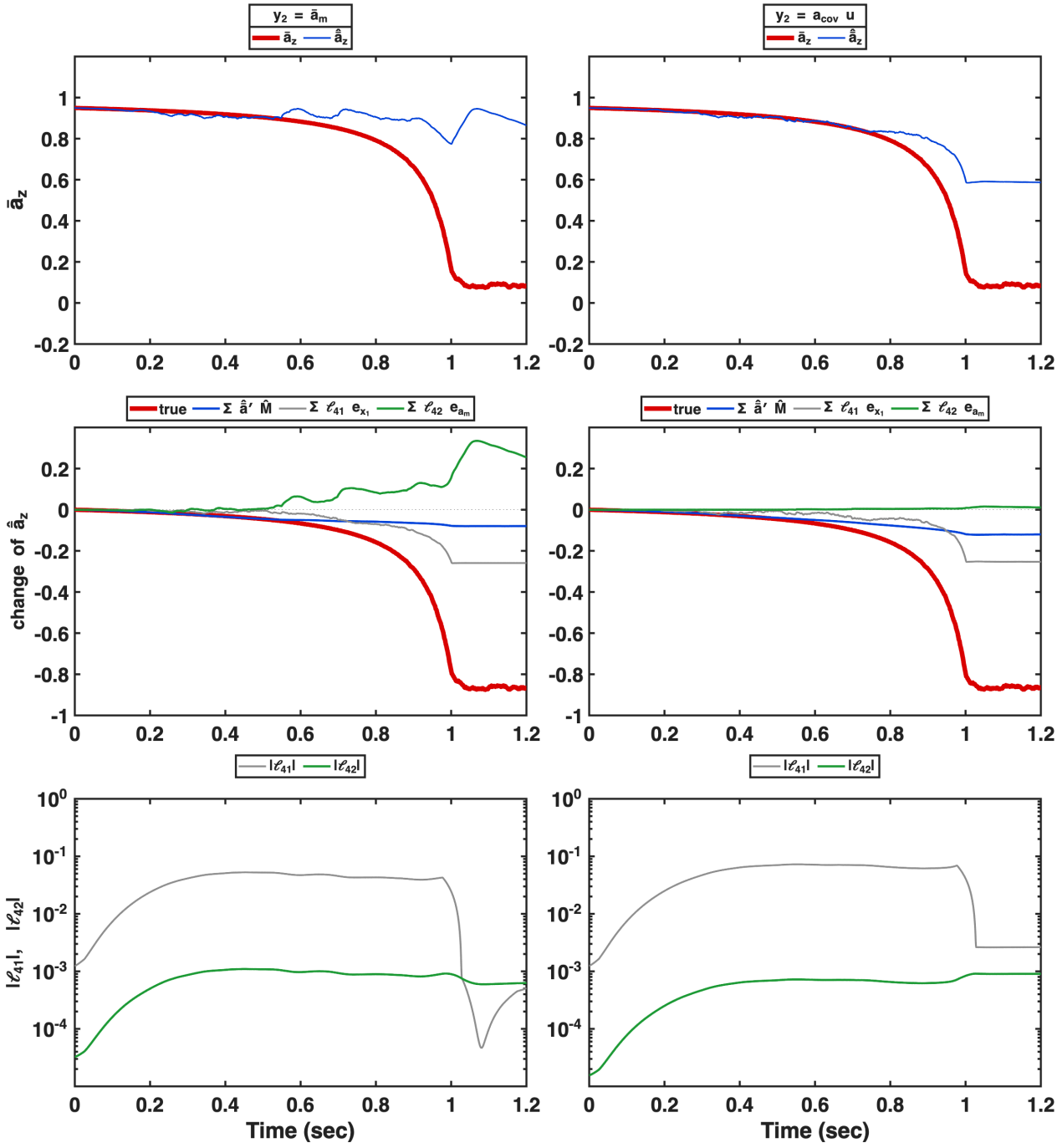
$$y_2[k] = \bar{a}_m[k] - (1 - a_{\text{cov}})\,\bar{a}_m[k-1] = a_{\text{cov}}\,u[k]$$

$$u[k] = \frac{\delta \bar{x}_r^2[k] - C_n\,\sigma_n^2}{C_T\,\kappa_T}$$

Fig. 4.1 Second output $y_2 = \bar{a}_m$ vs $y_2 = a_{\mathrm{cov}}\, u$

z axis; 22.2 → 1.10 R in 1 s, then held 0.2 s; plane wall; $q_{55} = 0$ in both columns; rows: $\bar{a}_z$; change of $\hat{\bar{a}}_z$ from $k = 0$ (true, $\Sigma\, \hat{a}' \hat{M}$, $\Sigma\, \ell_{41} e_{x_1}$, $\Sigma\, \ell_{42} e_{a_m}$); $|\ell_{41}|$, $|\ell_{42}|$ (50 ms mean, log scale); $\lambda_c = 0.4$, seed 1

$$\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\, \hat{M}[k] + \ell_{41}[k]\, e_{x_1}[k] + \ell_{42}[k]\, e_{a_m}[k]$$



5 Process noise of $\bar{a}'$

$\bar{a}'$ as a state variable

$$\bar{a}'[k+1] = \bar{a}'[k] + w_{\bar{a}'}[k]$$

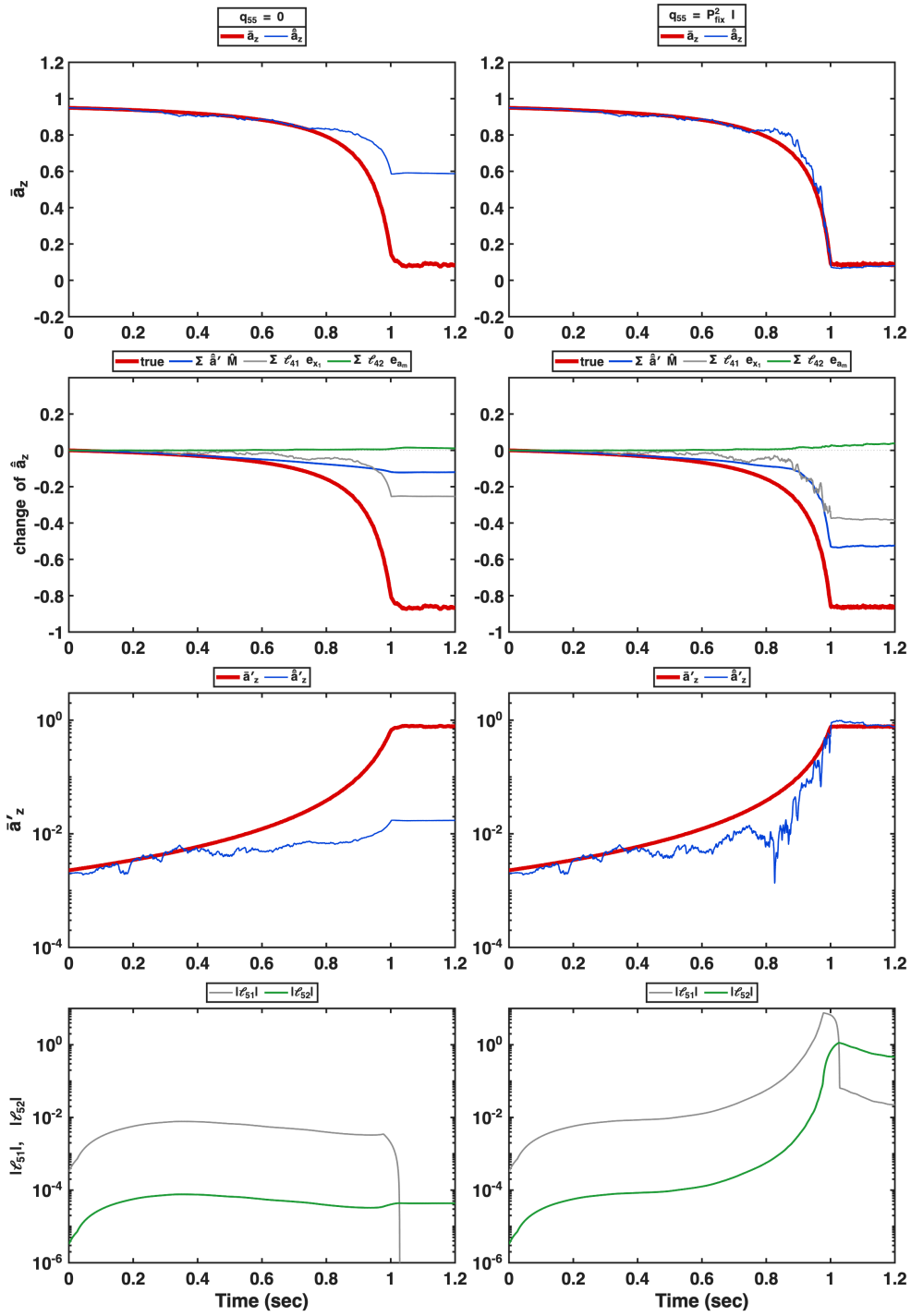
$$q_{55}[k] = 0 \quad \text{or} \quad p_{\text{fix}}^2 \, I[k]$$

$$\sqrt{p_{\text{fix}}} = 0.4 \left(1.1/h_d[k]\right)^2$$

Fig. 5.1 Process noise of $\bar{a}'$: $q_{55} = 0$ vs $q_{55} = p_{\text{fix}}^2\, I$

z axis; 22.2 → 1.10 R in 1 s, then held 0.2 s; plane wall; $y_2 = a_{\text{cov}}\, u$ in both columns; right column = left column of Fig. 2.1; rows: $\bar{a}_z$; change of $\hat{\bar{a}}_z$ from $k = 0$; $\bar{a}'_z$ (log scale); $|\ell_{51}|$, $|\ell_{52}|$ (50 ms mean, log scale); $\lambda_c = 0.4$, seed 1

$$\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\, \hat{M}[k] + \ell_{41}[k]\, e_{x_1}[k] + \ell_{42}[k]\, e_{a_m}[k]$$



Part III Regulation of the motion gain without the limit on $\Delta\bar{x}_d$

6 Contact with the wall, $\tau_a = 0.1$ s

Regulation law

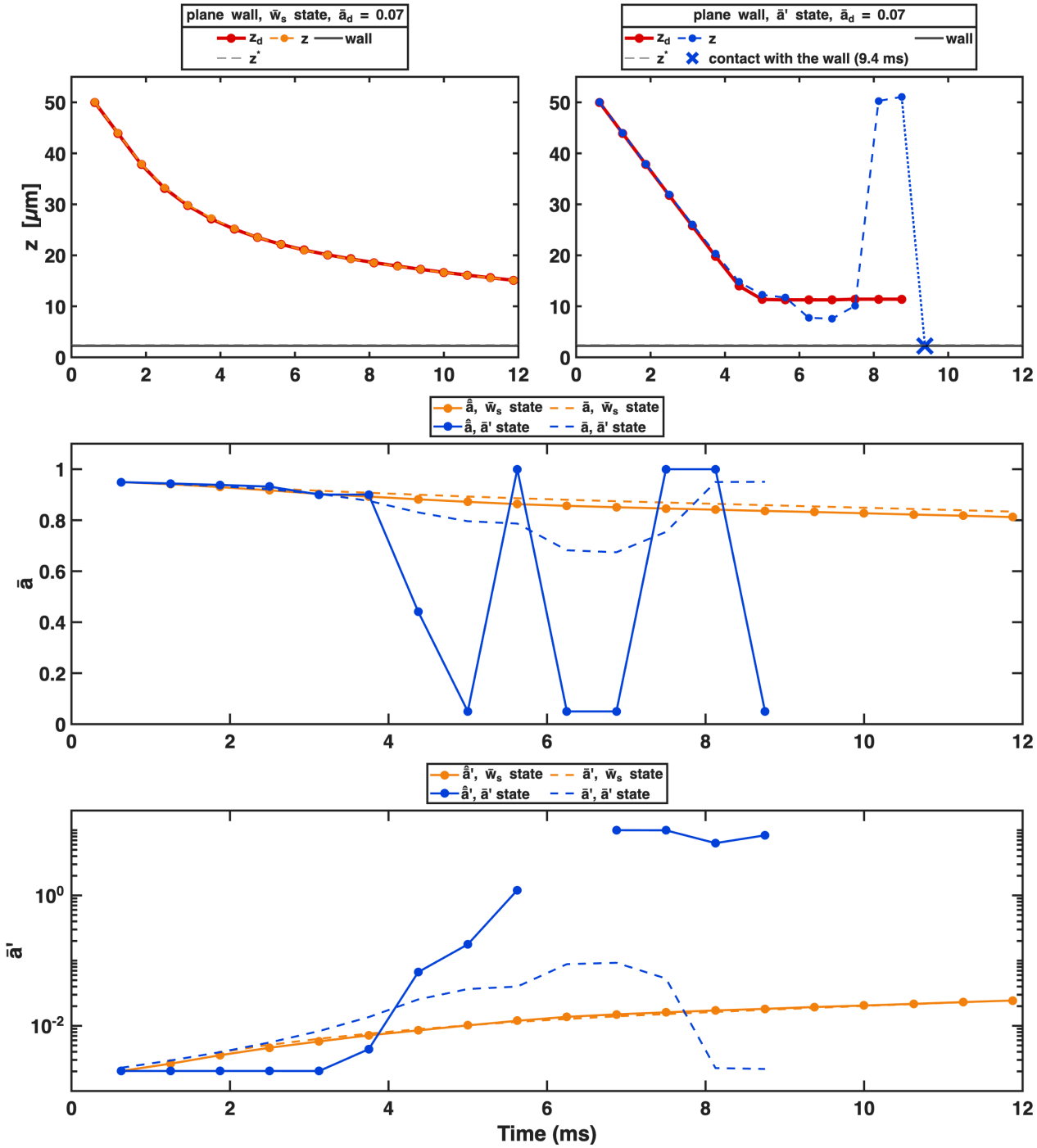
$$\Delta\bar{x}_d[k] = (1 - \lambda_a)\,\frac{\bar{a}_d - \hat{\bar{a}}[k]}{\hat{\bar{a}}'[k]}$$

$$\bar{x}_d[k+1] = \bar{x}_d[k] + \Delta\bar{x}_d[k]$$

$$\lambda_a = 1 - T_s/\tau_a$$

Fig. 6.1 $\tau_a = 0.1$ **s on the plane wall, first 12 ms**

z axis; start parked at 22.2 R; $\bar{a}_d = 0.07$; left: wall position $\bar{w}_s$ as a state variable; right: $\bar{a}'$ as a state variable; rows: z_d, z and z^* with the wall, $\bar{a}_z$, $\bar{a}'_z$ (log scale); one marker = one sampling period T_s; $\lambda_c = 0.4$, seed 1



7 Smallest τ_a without contact with the wall

Fig. 7.1 $\tau_a = 2$ s on the plane wall

z axis; start parked at 22.2 R, run 6 s; $\bar{a}_d = 0.07$; left: wall position $\bar{w}_s$ as a state variable; right: $\bar{a}'$ as a state variable; rows: z_d, z and z^* (log scale), $\bar{a}_z$, $\bar{a}'_z$ (log scale), δz_m; $\lambda_c = 0.4$, seed 1

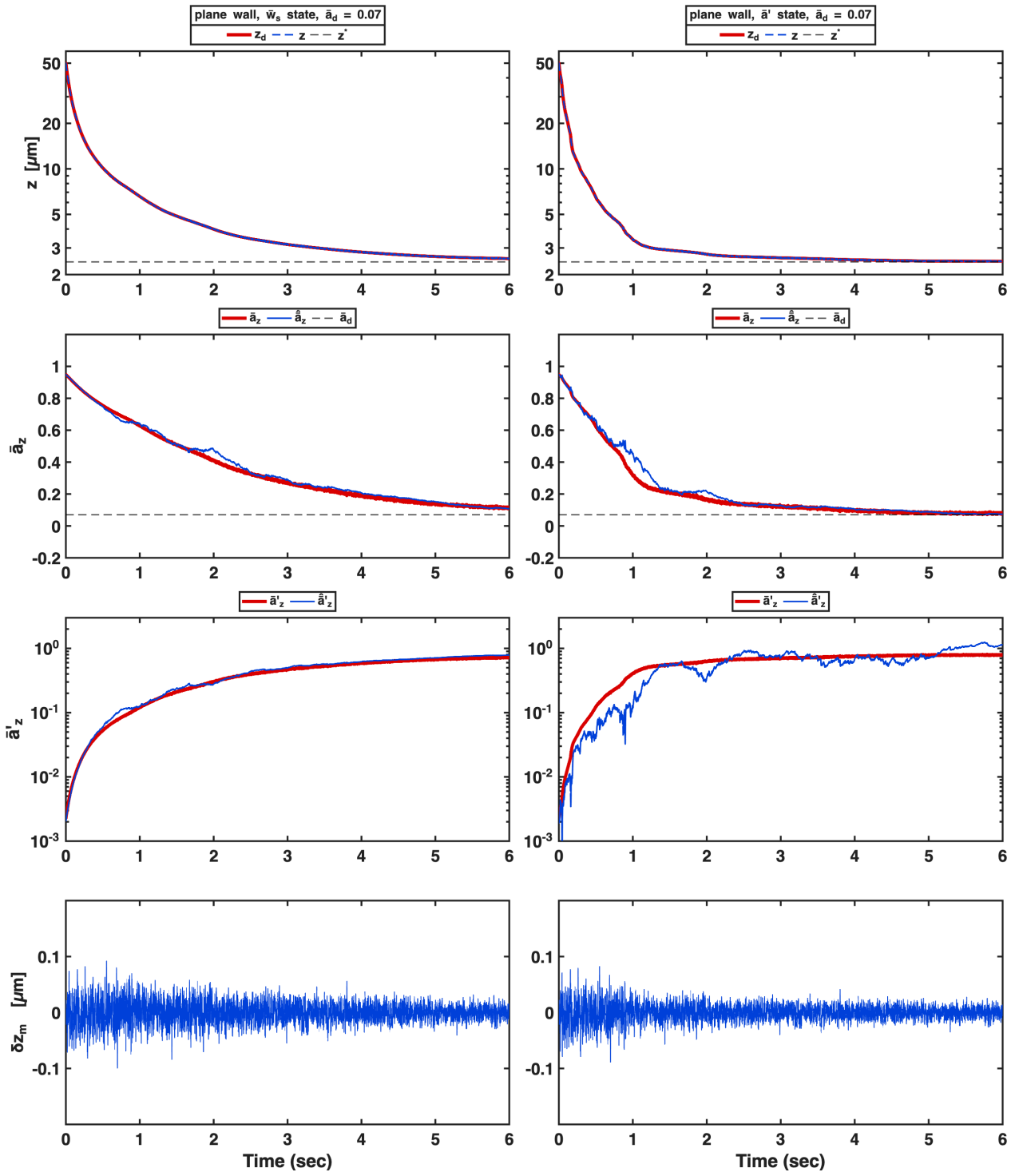


Fig. 7.2 $\bar{a}_z$ with $\tau_a = 2$ s on the plane wall

settings of Fig. 7.1, run 40 s; left: wall position $\bar{w}_s$ as a state variable; right: $\bar{a}'$ as a state variable; rows: $\bar{a}_z$ over 0–10 s; $\bar{a}_z$ over 10–20 s with the 1 s mean of the true $\bar{a}_z$; seed 1

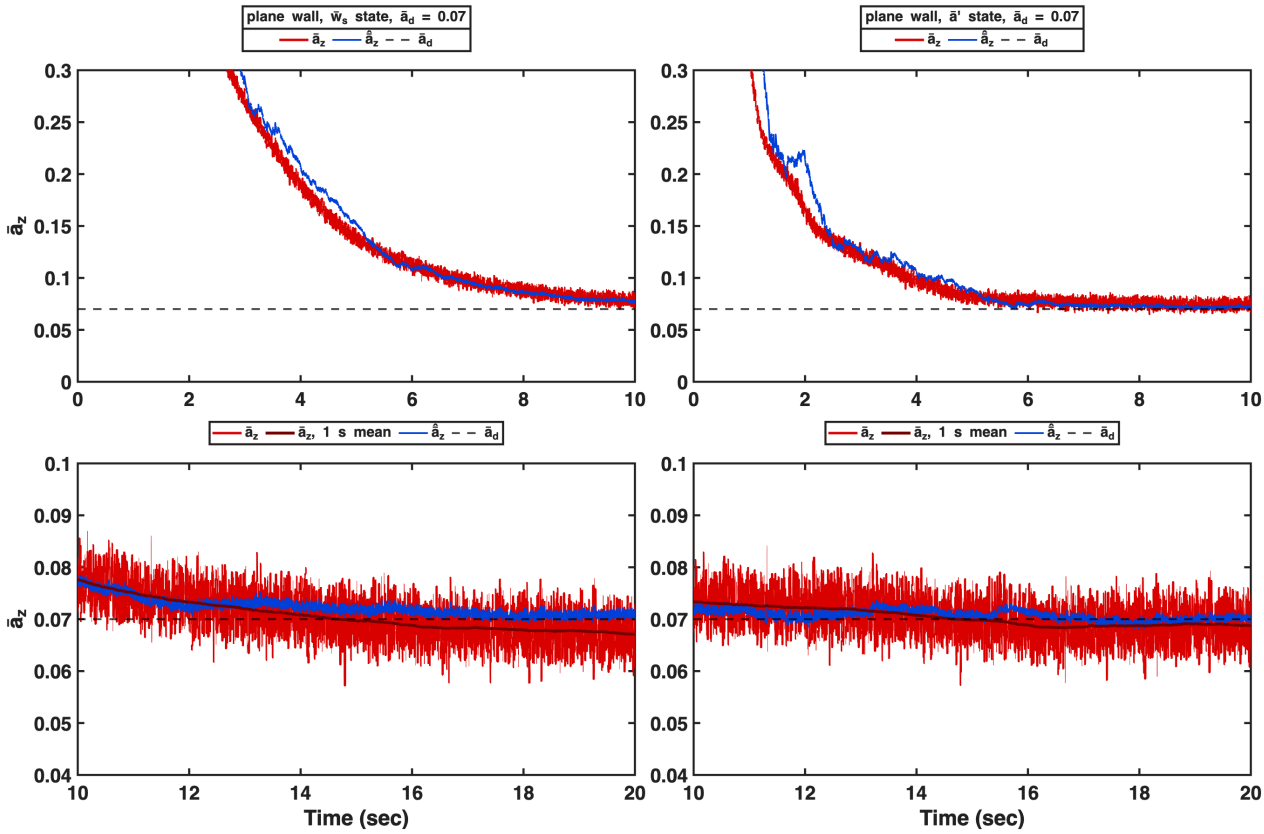


Fig. 7.3 $|\bar{a}_z - \bar{a}_d|$ **with** $\tau_a = 2$ **s on the plane wall**

settings of Fig. 7.1, run 40 s; left: wall position $\bar{w}_s$ as a state variable; right: $\bar{a}'$ as a state variable; rows: $|\bar{a}_z - \bar{a}_d|$, $|\hat{\bar{a}}_z - \bar{a}_d|$ and e^{-t/τ_a} (log scale), $\bar{a}'_z$ (log scale); seed 1

$$\bar{a}_d - \bar{a}[k+1] = \lambda_a \left(\bar{a}_d - \bar{a}[k]\right)$$

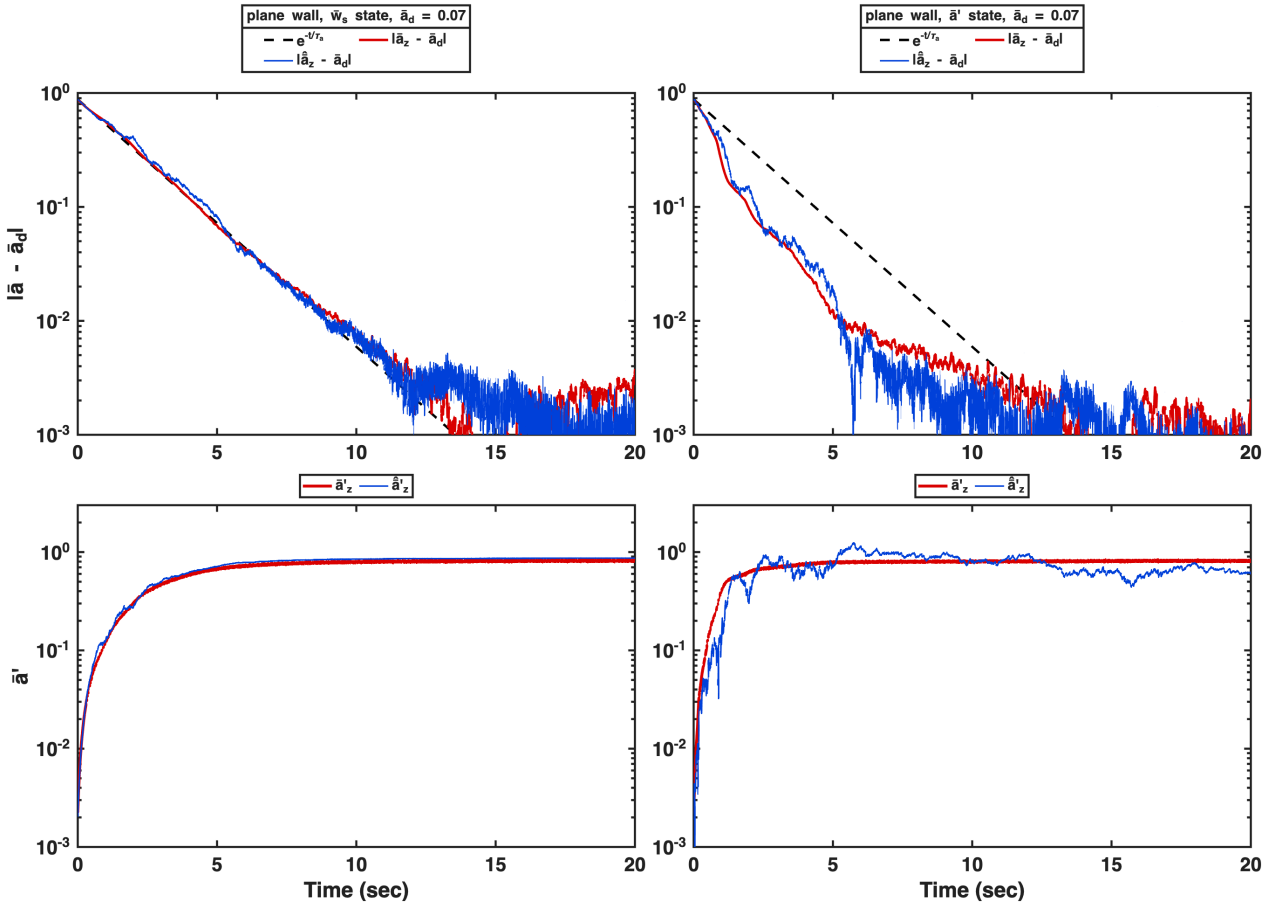


Fig. 7.4 $\tau_a = 2$ s on the cell wall

z axis; start parked at 22.2 R, run 6 s; $\bar{a}_d = 0.55$; cell wall = plane polynomial slid by 1.05 R, plant only; estimators started from the plane; left: wall position $\bar{w}_s$ as a state variable; right: $\bar{a}'$ as a state variable; rows as in Fig. 7.1; $\lambda_c = 0.4$, seed 1

